

- 6 Let μ be the Lebesgue measure on $\mathbb{R}$ and let $E \subset \mathbb{R}$ be a Lebesgue measurable set with $\mu(E) > 0$. We want to show that the set $E - E = \{x - y : x, y \in E\}$ contains an open interval centered at the origin. Since $\mu(E) > 0$ then we can write $\mu(E) = \sum_{n=-\infty}^{\infty} \mu(E \cap (n, n+1))$ and be assured that there exists n' such that $\mu(E \cap [n', n'+1]) > 0$. If we consider $E - n$ then we can apply problem 5 and find an open interval $I \subseteq (n, n+1)$ such that $\mu(E \cap I) > r\mu(I)$ for all $r \in (0, 1)$. Note that this implies that $\mu(E \cap I) = \mu(I)$. Therefore we know for arbitrary $\epsilon > 0$ that $\mu(E \cap [n, n+1] \Delta I) < \epsilon$. We claim that the function $\phi(x) = \mu((E \cap I) \cap (E \cap I + x))$ is continuous. Note that for the function $\psi(x) = \mu(E \cap I \cap I + x)$ we can compute it exactly, given that there exists $n < a < b < n+1$ such that $I = (a, b)$ then we have that $\psi(x) - \psi(y) = \mu(E \cap I \cap I + x) - \mu(E \cap I \cap I + y) = b - a - |x| + b - a - |y| = |y| - |x|$. where $x, y \in (a-b, b-a)$. Furthermore, if we consider $\phi(x) - \psi(x) = \mu((E \cap I) \cap (E \cap I + x)) - \mu(I \cap I + x)$, since we have that

$$\begin{aligned} \mu((E \cap I) \cap (E \cap I + x) \Delta (I \cap I + x)) &= \mu((I \cap I + x) \setminus (E \cap I) \cap (E \cap I + x)) \\ &= \mu(\end{aligned}$$

I didn't have complete time, but if I could suss out the convergence of the ϕ then from that one can take an ϵ neighborhood around some ψ which is non-zero, then from there you can look at the set it gives out in ψ^{-1} . Since this will be an open set you can then subtract every element by every element within the delta neighborhood, which has the effect of translating the neighborhood to have an edge on the origin, include the origin, then reflect it across the axis to reflect sign flips of the elements subtracting.

- 7 Let $\mathcal{F}$ be a countable family of Lebesgue measurable functions, E be a Lebesgue measurable set with $\mu(E) < \infty$ with the property for each $x \in E$ there exists $M_x < \infty$ such that for all $f \in \mathcal{F}$, $|f(x)| \leq M_x$, and $\epsilon > 0$. We want to construct a closed set $F \subset E$ such that there exists $M > 0$ that for all $x \in F$, $f \in \mathcal{F}$, $f(x) \leq M$ with $\mu(E \cap F^c) \leq \epsilon$. Let $f' = \sup_{f \in \mathcal{F}} f$. Note that f' is a Lebesgue measurable function. Note that since every point $x \in E$ every function in $\mathcal{F}$ has a finite pointwise bound. Therefore f' takes on $\mathbb{R}$ values on E . Therefore $f'^{-1}((-\infty, \infty)) \cap E = E$, as every point in E is eventually included. Therefore there exists $\infty > M > 0$ such that $\mu(f'^{-1}([-M, M]) \cap E) \geq \mu(E) - \epsilon/2$. Furthermore by the inner regularity of μ we have that there exists $F \subset f'^{-1}([-M, M]) \cap E$ such that $\mu(F) \geq \mu(f'^{-1}([-M, M]) \cap E) - \epsilon/2$. Therefore we have that

$$\begin{aligned} \mu(E) &= \mu(F) + \mu(E \cap F^c) \\ &\geq \mu(E) - \epsilon + \mu(E \cap F^c) \\ \epsilon &\geq \mu(E \cap F^c) \end{aligned}$$

Additionally, by taking the preimage of $[-M, M]$ of the sup of the functions of $\mathcal{F}$ the uniform boundedness is implied.

- 8 Let E be a non-Lebesgue measurable set and let A be a Lebesgue measurable set of measure 0. We claim that $E \cap A^c$ is not Lebesgue measurable. Suppose not. Then $E \cap A^c$ is Lebesgue measurable. Additionally, since E is not Lebesgue measurable then

there exists $C \subset \mathbb{R}$ such that $\mu^*(C) < \mu^*(E \cap C) + \mu^*(E^c \cap C)$, where μ^* is the outer Lebesgue measure. Furthermore, we note that $E \cap A$ must be Lebesgue measurable since $E \cap A \subseteq A$, and therefore $0 \leq \mu^*(E \cap A) \leq \mu(A) = 0$, which by the definition of the completeness of measure spaces implies that $E \cap A$ is Lebesgue measurable (a similar argument works for $\mu^*(E^c \cap A)$). Since $A \cap E, A \cap E^c, A^c \cap E$ are all measurable then their union is measurable, $A \cup E$, and the complement of their union is measurable, $A^c \cap E^c$.

Therefore we have that

$$\begin{aligned} \mu^*(C) &< \mu^*(C \cap E) + \mu^*(C \cap E^c) \\ &= \mu^*(C \cap E \cap A) + \mu^*(C \cap E \cap A^c) + \mu^*(C \cap E^c \cap A) + \mu^*(C \cap E^c \cap A^c) \\ &= \mu^*(C \cap (E \cup A)) + \mu^*(C \cap E^c \cap A^c) \\ &= \mu^*c(C). \end{aligned}$$

Note that this is a contradiction.

- 9 Let $A_0 \subset A_1 \subset \dots \subset A_n \subset \dots [0, 1]$. We want to show that $\lim \mu^*(A_n) = \mu^*(\bigcup_{n=1}^{\infty} A_n)$. Note by the outer regularity of μ^* there exists B_n open such that $A_n \subset B_n$ and $\mu^*(B_n) \leq \mu^*(A_n) + 2^{-n}$, and $C_n = \bigcap_{m=n}^{\infty} B_m$. Since each C_n is a measurable set then we have that $\lim \mu(C_n) = \mu(\bigcup_{n=1}^{\infty} C_n)$. Note that $\mu(\bigcup_{n=1}^{\infty} C_n) \leq \sum_{n=1}^{\infty} \mu(C_n)$. Furthermore, if we consider that (I ran out of time, sorry).
- 10 Let μ be the Lebesgue measure on $\mathbb{R}$, and let $0 \leq a < b \leq 1$. We will construct an open set $U \subset [0, 1]$ such that $\mu(A) = a, \mu(B) = b$. If we restrict ourselves to the interval $[0, b]$, and choose an $\alpha \geq 1$ such that $\frac{a}{b} < x', 1 - 2x'^{\alpha} - x', x' > 0$. Then we begin to construct the set C as follows: for C_1 , we remove the middle r portion of $[0, b]$ and we're left with $[0, \frac{r}{2}] \cup [\frac{r}{2} - b, b]$. Then for C_2 we remove the middle $r(\alpha + 1)$ terms from each remaining interval, dividing them in two. At the n th step we remove $2^{n-1}r^{1+\alpha(n-1)}$ from the collective remaining intervals. The final set we end up with is $C = \bigcap_{j=1}^{\infty} C_j$. Note that C is closed. For our desired set, we take $U = C^c$. Note that $\mu(U) = b - \sum_{i=1}^{\infty} 2^{i-1}r^{1+\alpha(i-1)} = b \frac{1-2r^{\alpha}-r}{1-2r^{\alpha}}$. Note that since $\frac{a}{b} < 1 - 2x'^{\alpha} - x'$ then we have that the numerator is not 0, and ensure that the function will be positive. Then we can choose a positive r such that $a = \frac{1-2r^{\alpha}-r}{1-2r^{\alpha}}$, since the area function is continuous on $[0, x']$, and by the intermediate value theorem since $r = 0$ corresponds to 1 and $r = x'$ corresponds to 0, and in the region our function is positive. Therefore $\mu(U) = b \frac{a}{b} = a$. Furthermore, $\mu(\bar{U}) = b$ since U is the interval excluding well chosen limit points.